Introduction to Optical Transmitters Tx1

Lecture Series

University Program
Photonics Curriculum Version 8.0

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Module Prerequisites

- Introduction to Fiber-Optic Communications I & II
- Recommended - Fibers I

Module Objectives

- Semiconductor lasers as sources for optical transmitters
- How a laser works
  - Functional view: gain medium, energy pump, cavity, losses
  - Basic structure: Fabry-Perot laser
  - Optical absorption and emission processes
- Gain Curve, Lasing conditions (gain and phase)
- Rate Equations, Dynamic Effects of Lasers
- Introduction to DFB Lasers

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Sources for Optical Transmitters

- Many types of optical sources are available
  - Light Emitting Diodes (LEDs)
  - Solid state lasers
  - Gas lasers
  - Semiconductor lasers
  - Fiber lasers
- Semiconductor Lasers are preferred
  - Powered by electrical energy
  - Directly converts electrical signals to optical signals
  - Generate coherent light source unlike LEDs

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Semiconductor Laser Features

- High modulation bandwidth (> 10 Gbit/s)
- Small size
  - Packaged: ~ 2 × 1 × 1 cm
  - Unpackaged: ~ grain of salt, 0.5mm × 200μm × 100μm
- Intense single spatial mode
- Energy efficient
- Narrow spectral linewidth
- Can be single longitudinal mode (monochromatic)
- Reliable operation
- Can be integrated

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What does LASER stand for?

LASER is an acronym for:

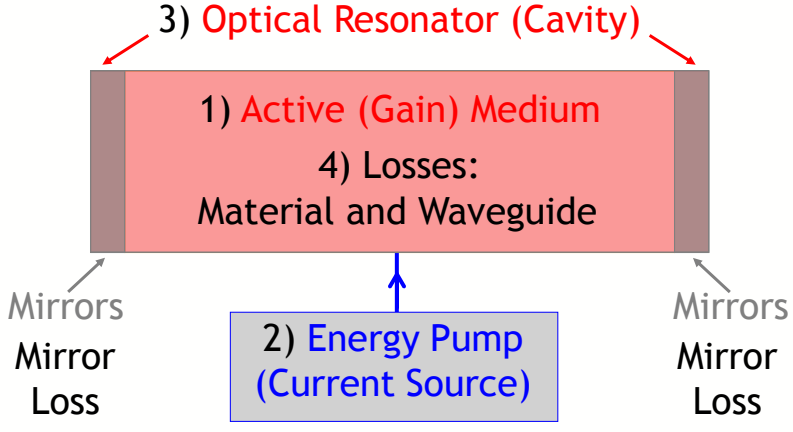
- **L**IGHT
- **A**MPLIFICATION by
- **S**TIMULATED
- **E**MISSION of
- **R**ADIATION

- How does it work?
- Why amplification?
- What is stimulated emission?

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How does a Laser work?

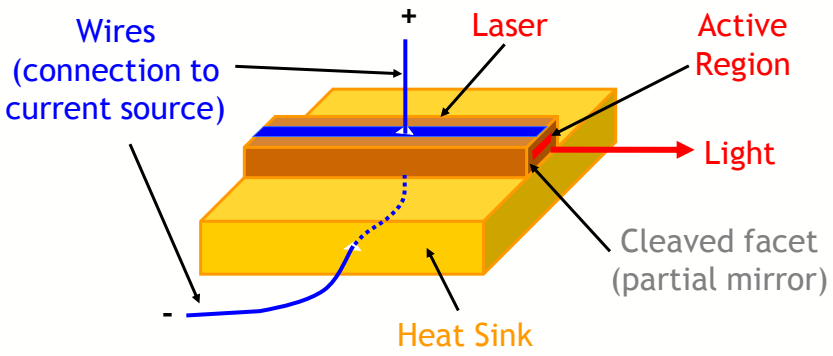
A functional view of a laser: 4 main parts



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Basic Laser Structure

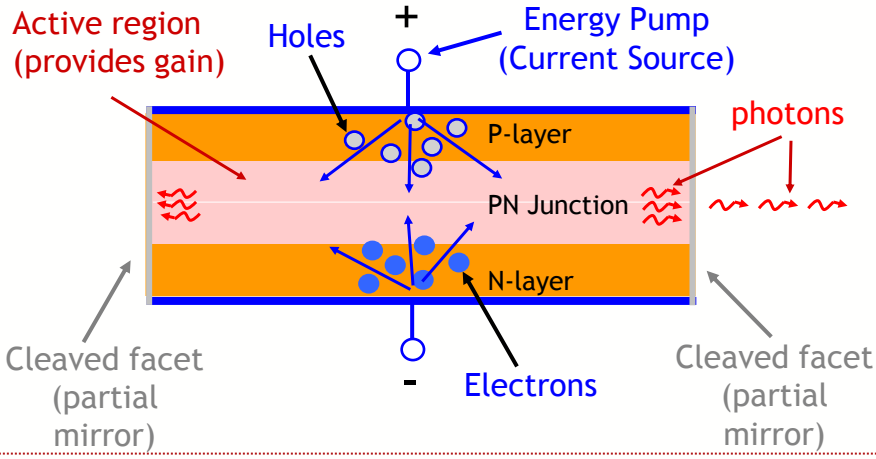
Fabry-Perot Laser



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A Basic Laser Structure

Fabry-Perot Laser (longitudinal section)

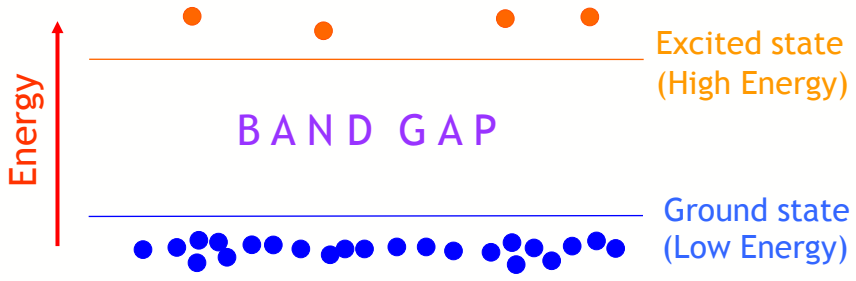


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How does a Laser produce Light?

- Need to consider three optical transition processes:
  - Absorption
  - Spontaneous emission
  - Stimulated emission

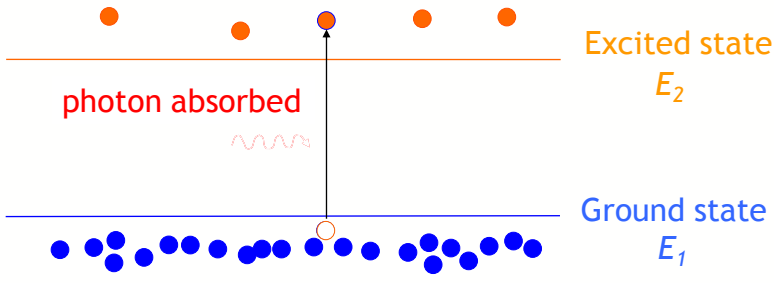
Energy level diagrams for electrons in the active medium



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Optical Absorption Process

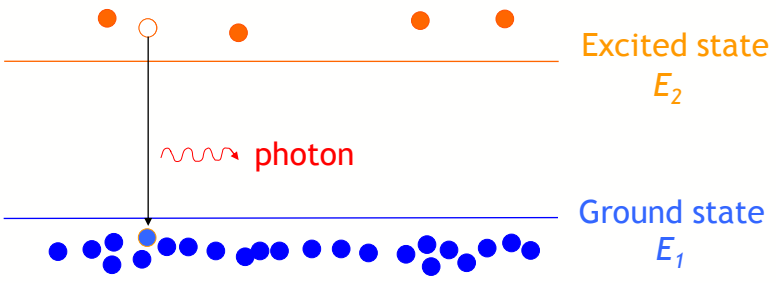
- Absorption: a photon with energy > (E_2 - E_1)
- The photon's energy can be absorbed by an electron in state E_1, thus exciting it to state E_2



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Optical Emission Processes

- Spontaneous Emission: electron in excited state E_2 can spontaneously decay to state E_1
- A photon with energy $hf > (E_2 - E_1)$ is emitted

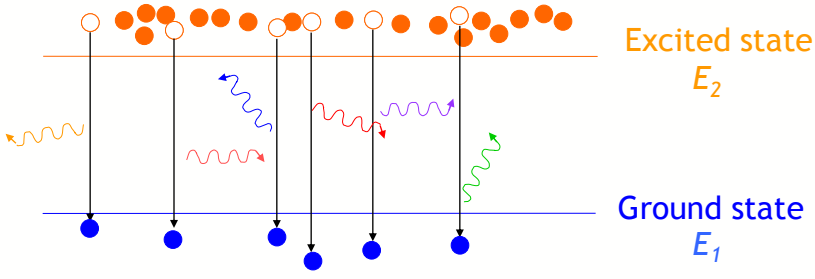


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Optical Emission Processes

Light produced by spontaneous emission:

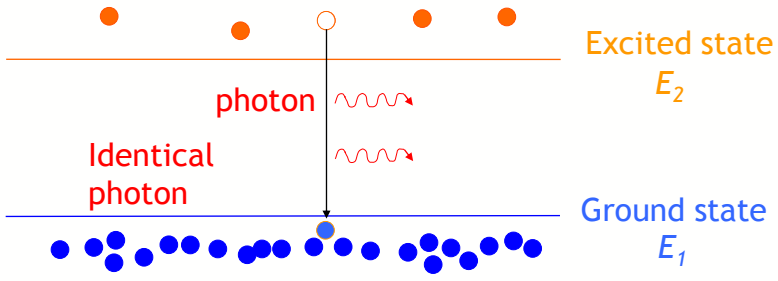
- Random propagation direction
- Random phase
- Random frequency
- is incoherent (broad linewidth)



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Optical Emission Processes

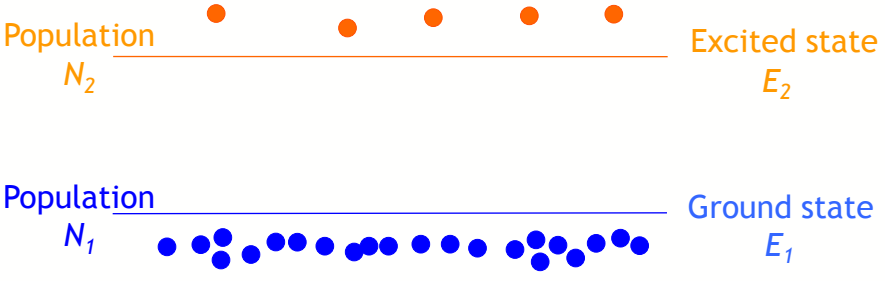
- Stimulated Emission
- A photon with energy > (E_2 - E_1): triggers transition of an excited electron → identical photon is emitted
- Produced light is coherent (desirable)



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Optical Transition Processes

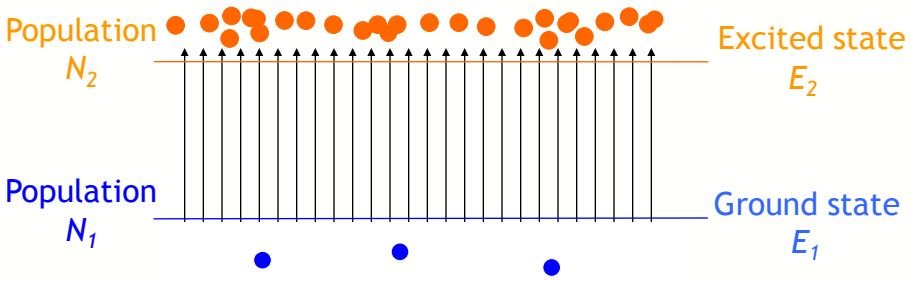
- All three processes related (Einstein Relations)
- At thermal equilibrium, absorption = emission
- Electrons mostly in state E_1: (N_2 << N_1)
- Spontaneous emission dominates



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Population Inversion

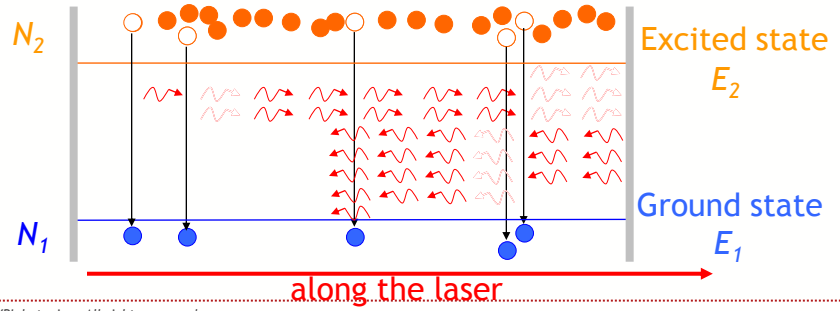
- Stimulated emission – coherent light, desirable
- Need electrons to be mostly in state E_2
- Population inversion achieved by electrical pumping
- N_2 >> N_1, stimulated emission dominates



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Optical Cavity and Feedback

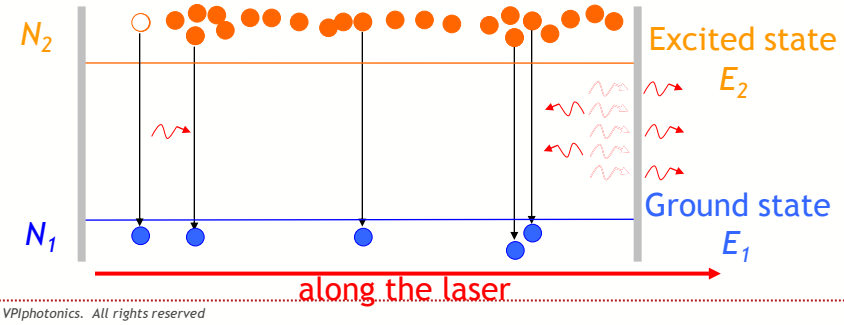
- Light amplification by stimulated emission…
  is not strong, especially if the active region is short
- Optical cavity provides feedback into active region
- Light is reflected repeatedly and greatly amplified



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Optical Losses and Laser Output

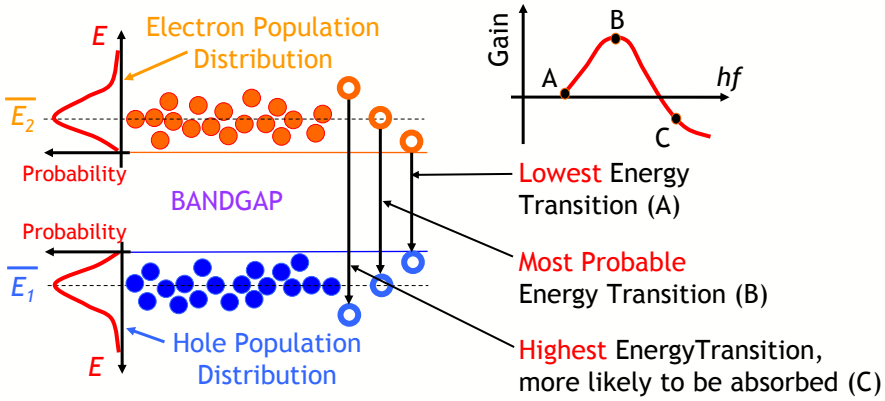
- One mirror is partially transmitting, to get output
- Photons transmitted out are lost (mirror loss)
- Other losses: scattering in the material, nonradiative processes



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Laser Spectrum (Gain vs. λ Curve)

- Laser oscillation (lasing) occurs over a range of λ
- Due to distribution of E_2 and E_1 around mean values



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Lasing Conditions

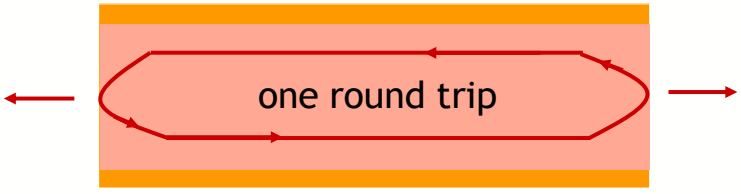
For lasing to initiate in a cavity, two conditions need to be satisfied:

- Gain condition:

  The electric field of the light, after completing one round trip inside the cavity, should have the same amplitude

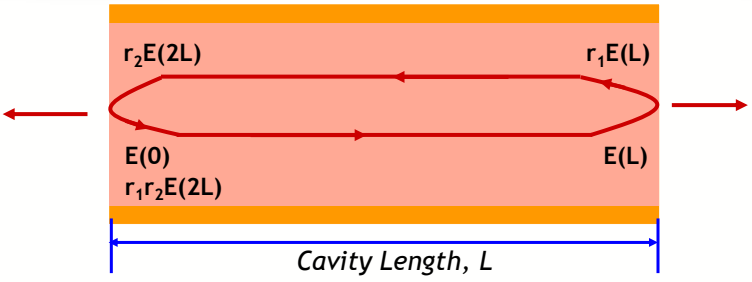
- Phase condition:

  The electric field of the light, after completing one round trip inside the cavity, should have the same phase



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Gain Condition



Gain Condition: $E(t,0) = E(t,2L)$

$E(t,0) = A\ exp(j\omega t);\ E(t,\ 2L) = Ar_1r_2exp[2L(g-\alpha i)]exp[j(\omega t-2\beta L)]$

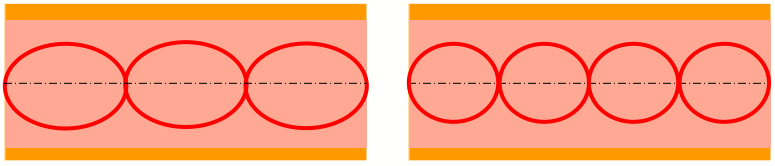
To satisfy the Gain Condition: $r_1r_2exp[(g-\ \alpha_i)2L] = 1$

The threshold gain: $g_{th} = \alpha_i + \frac{1}{2L} ln \frac{1}{r_1r_2}$

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Phase Condition

Integral number of cycles must fit within the cavity



Phase: $exp(-2j\beta L) = exp\left(-\frac{4j\pi nL}{\lambda}\right) = 1, \quad \frac{4j\pi nL}{\lambda} = 2m\pi$

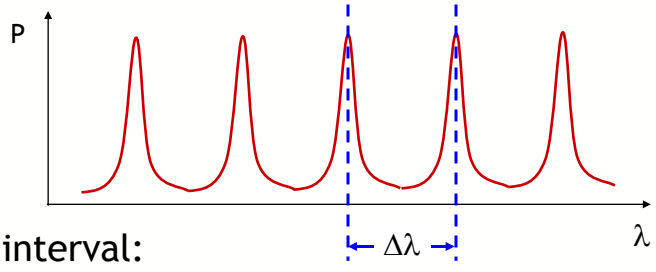
Resonant cavity – discrete set of spectral lines



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Phase Condition

Discrete modes that can be supported by the cavity are called longitudinal modes



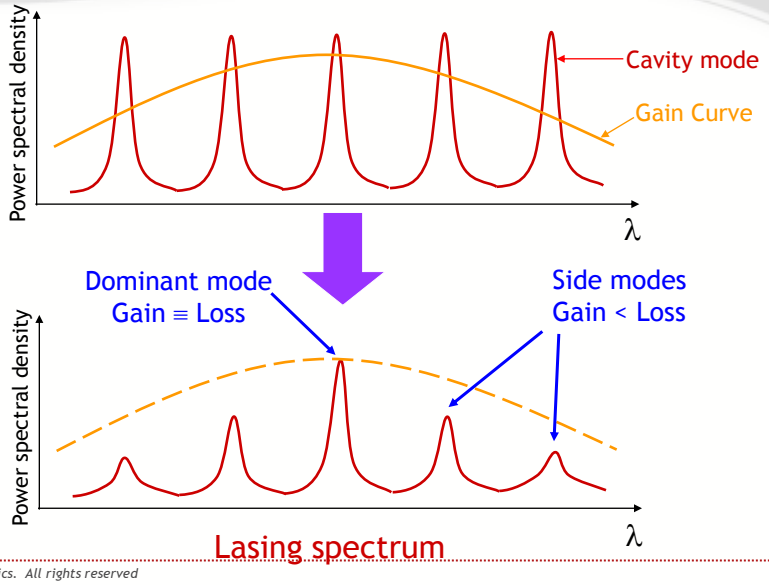
Mode interval:

Phase condition: $\frac{2nL}{\lambda} = m$

Thus: $\Delta\lambda = \frac{\lambda^2}{2nL}$

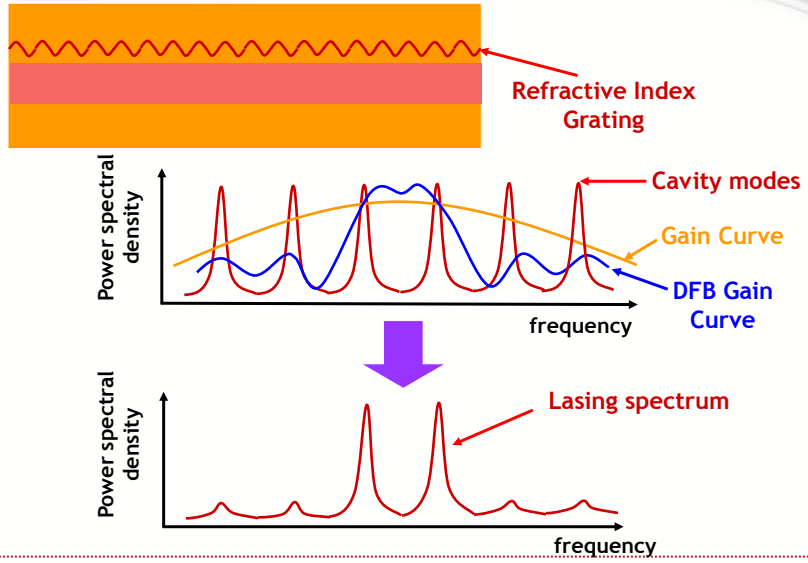
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Laser Spectrum: Gain & Phase conditions met



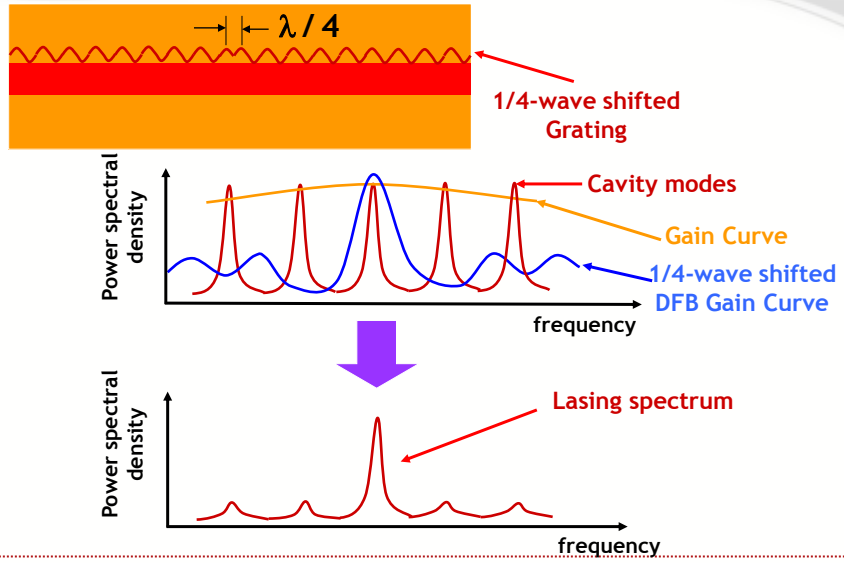
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Another Laser Structure: DFB Laser



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Another Laser Structure: DFB Laser



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Rate Equation

Rate equation describe the dynamic interaction between excited electrons and photons

Change in carrier density:

$$\frac{dn}{dt} = \frac{I}{qV} - \frac{n}{\tau} - v_g g s$$

Contribute from Current

Decay of electrons

Usage of electron by Stimulated emission

Change in photon density:

$$\frac{ds}{dt} = \Gamma v_g a(n-n_0)s - \frac{s}{\tau_p} + \Gamma\beta\frac{n}{\tau}$$

Photon generated by stimulated emission

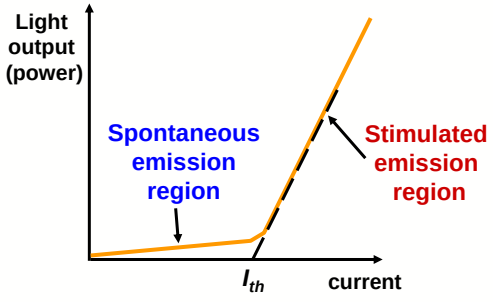
Decay of photon

Photon generated by Spontaneous emission

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Light-Current Characteristics

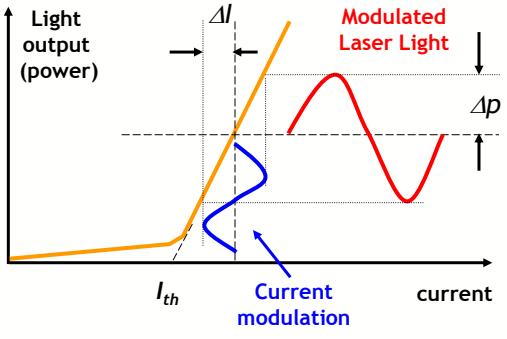
- The *L-I* curve: output light power vs. input current
- Diode characteristics, threshold current I_{th}
- For $I > I_{th}$, light power increases linearly with I



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Direct-Current Modulation

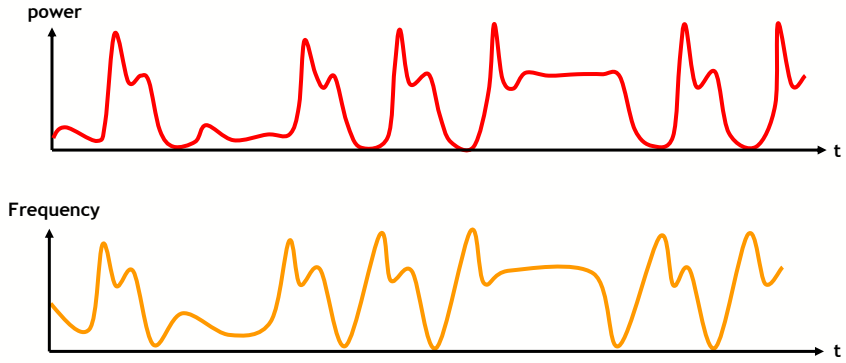
The information is encoded on semiconductor lasers by current modulation



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Laser Dynamics: Chirp

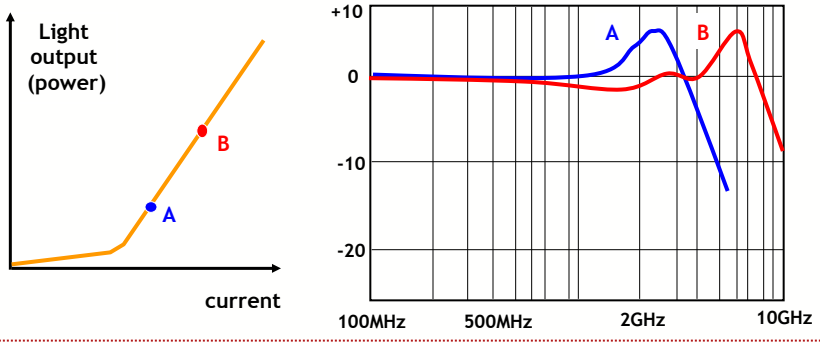
- Chirping of the laser output signal (pulses)
- Instantaneous frequency of the signal changes
- Leads to increased dispersion (broadening)



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Laser Dynamics: Modulation Bandwidth

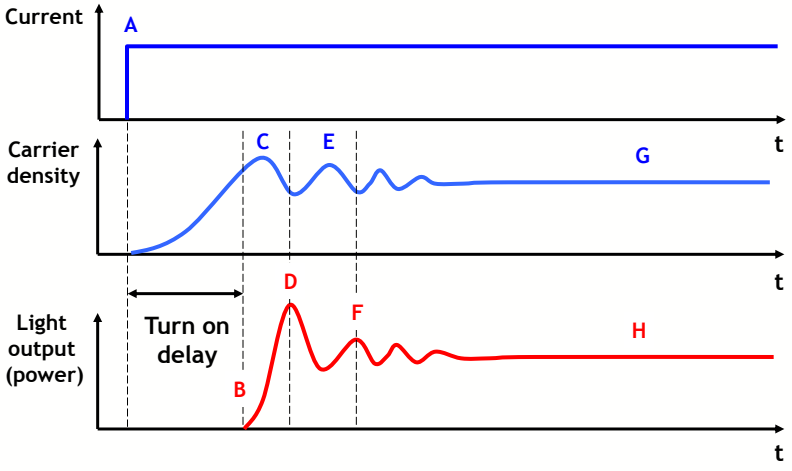
- Determines maximum direct modulation speed
- Increases with increasing drive (bias) current
- Within practical limits, get multi-GHz modulation



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Laser Dynamics: Turn On Delay

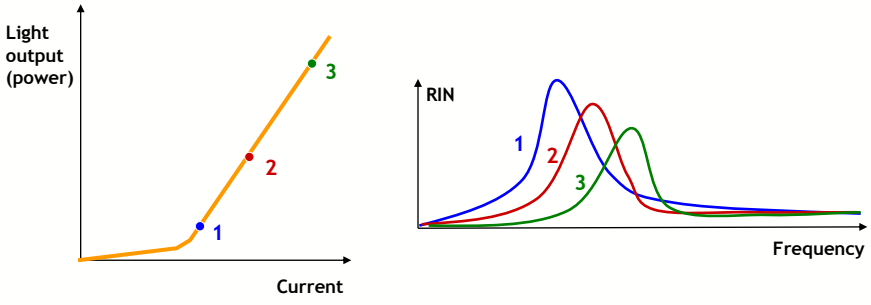
Delay between injection of current and generation of light



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Laser Dynamics: Relative Intensity Noise

Frequency shape of RIN depends on laser driving conditions



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Summary

- Semiconductor lasers optical transmitters
  - Gain medium, energy pump, cavity, losses
  - Basic structure: Fabry-Perot laser
  - Optical absorption and emission processes
  - Einstein Relations
- Gain Curve, Lasing conditions (gain and phase)
- Introduction to Dynamic Effects of Lasers
  - Rate equations, *L-I* curve, direct modulation, chirp
  - Modulation bandwidth, turn on delay, RIN

Proceed with the *Interactive Learning Module*

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